SI Session 1 Review

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1 Problems

1.1 Separation of Variables

1.1.1 Problem 1

Solve for y:

$$\frac{dy}{dx} = \frac{x^3}{1+x^4}(1+y^2)$$

1.1.2 Problem 2

$$\frac{dy}{dx} = \frac{x}{x^2+1}\frac{1}{y\ln(y)}$$

1.2 Homogeneous

1.2.1 Problem 3

Eliminate fractions

$$\frac{dy}{dx} = \frac{2xy}{x^2-y^2}$$

1.2.2 Problem 4

$$\frac{dy}{dx} = \frac{x^2+y^2}{xy}$$

1.3 Exact

1.3.1 Problem 5

$$(2xy + y^2 + \frac{y}{x} + \tan^{-1}(y))dx + (x^2 + 2xy + \ln(x) + \frac{x}{1+y^2})dy = 0$$

1.4 Linear

1.4.1 Problem 6

$$xy' + y = x^4$$

1.4.2 Problem 7

$$y' + \frac{2x}{1+x^2}y = \frac{e^x x}{1+x^2}$$